

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and data scientist specializing in data quality, statistical diagnostics, and root-cause analysis for large, noisy, and dependent datasets. Builds reproducible Python/PyTorch workflows for data curation, anomaly investigation, simulation, and ongoing analytical monitoring. Experienced in translating ambiguous financial and scientific questions into mathematically defined analyses and clearly communicating findings to technical and business stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Feb 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), R, SQL, C++/pybind11 integration, Bash, Linux/Unix, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analysis pipelines

Data Analysis and Monitoring: Data curation, aggregation, exploratory analysis, visualization, data validation, anomaly detection, root-cause analysis, statistical testing, regression, GLM/GBM, Monte Carlo simulation, model diagnostics, feature attribution, performance evaluation, and stakeholder reporting

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, converting a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled stopping decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to quantify Type I/II error, boundary-crossing probabilities, stopping behavior, uncertainty, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM analysis pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Analyzed performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

RESEARCH AND ANALYTICAL SYSTEMS

Ph.D. Dissertation Research - Data Quality, Scalable Analysis, and Statistical Diagnostics

Sep 2024 - Present

- Data Curation and Quality:** Built workflows for quality filtering, missingness tracking, time-specific coordinate offsets, nearest-neighbor center matching, and spatial-threshold regular-grid construction for large satellite datasets.
- Root-Cause Analysis:** Investigated scan-geometry and time-dependent alignment effects and designed scale- and frequency-resolved diagnostics to distinguish systematic acquisition artifacts from random noise, missingness, and statistical-model misspecification.
- Scalable Statistical Analysis:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, using ordered local conditioning to analyze up to 145,008 observations per day.
- Analysis Automation:** Implemented reusable Python/PyTorch pipelines for preprocessing, estimation, simulation, cached outputs, and restartable CPU/GPU and HPC jobs; integrated compiled C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Linear Algebra, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)